

Jamhuriya RMS

Database Structure Documentation

Research Management System (MERN Stack)
MongoDB + Mongoose ODM

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Database: rms | URI: mongodb://localhost:27017/rms
Repository: research-management-systems

1. Overview

Jamhuriya RMS uses MongoDB with 15 collections (Mongoose models). All collections include createdAt and updatedAt timestamps unless noted.

Collections: users, departments, proposals, ethicsapplications, projects, publications, grants, budgets, payments, purchaseorders, researchgroups, thesishgroups, conversations, notifications, repositoryitems.

2. User Roles

Role	Code	Access
Research Director	research_director	Institutional oversight, ethics approval, grants
Faculty Coordinator	faculty_coordinator	Faculty proposals, publication validation
Finance Officer	finance_officer	Budgets, payments, procurement
Researcher	researcher	Proposals, projects, publications, groups

3. Entity Relationships

- User !' Proposal (researcherId)
- User !' EthicsApplication (researcherId)
- Proposal !' EthicsApplication (1:1 via proposalId / ethicsApplicationId)
- Proposal !' Project (approved proposal creates project)
- User !' Project (researcherId = PI)
- Project !' Grant !' Budget !' Payment / PurchaseOrder
- User !' Publication (researcherId)
- ResearchGroup !' Conversation (groupId for group chat)
- ThesisGroup !' ResearchGroup (researchGroupId)
- User !' Notification (userId)
- User !' RepositoryItem (uploadedBy)
- Department !' ResearchGroup (departmentId)

4. Main Workflow

1. Researcher registers !' User (status: pending !' active after approval)
2. Researcher creates Proposal + EthicsApplication (draft)
3. Submit to Director !' status: submitted
4. Director reviews ethics + proposal !' approved
5. System creates Project from approved proposal
6. Researcher applies for Grant !' Budget allocated
7. Payments / PurchaseOrders drawn from Budget (director !' finance)
8. Publications submitted !' validated by coordinator/director
9. Files uploaded to Repository (PDF/CSV/Excel)
10. Collaboration via ResearchGroup + Conversation + Notifications

5. Collection Reference

1. users (User)

Authentication, roles, and researcher profiles.

Field	Type	Description
fullName	String	Required — display name
email	String	Required, unique, lowercase
password	String	Required, bcrypt hashed (select: false)
role	Enum	research_director faculty_coordinator finance_officer researcher
department	String	Required
rank	String	Required academic rank
researchInterests	String	Comma-separated interests
status	Enum	pending active rejected
isProtected	Boolean	Protected bootstrap/admin accounts
refreshToken	String	JWT refresh token (select: false)
createdAt / updatedAt	Date	Auto timestamps

2. departments (Department)

Faculty departments under Jamhuriya University.

Field	Type	Description
name	String	Required, unique
code	String	Required, unique (e.g. CS)
faculty	String	Faculty / kulliyad name
createdBy	ObjectId !' User	Who created the record
createdAt / updatedAt	Date	Auto timestamps

References: User (createdBy)

3. proposals (Proposal)

Research proposals submitted before project creation.

Field	Type	Description
title, abstract	String	Required
department, researchArea	String	Required
document	String	Uploaded file path (optional)
version	Number	Default 1
versionHistory	Array	Version snapshots with document + note
researcherId	ObjectId !' User	Proposal owner
ethicsApplicationId	ObjectId !' EthicsApplication	Linked ethics form
status	Enum	draft submitted under_review approved rejected revision_requested
requiresEthics	Boolean	Default true
ethicsStatus	Enum	not_required pending approved rejected revision_requested
ethicsComments	Array	role, comment, at
assignedReviewers	Array	userId, assignedBy, assignedAt
reviewerComments	Array	role, comment, at
submittedAt	Date	Submission timestamp
createdAt / updatedAt	Date	Auto timestamps

References: User (researcherId, reviewers), EthicsApplication

4. ethicsapplications (EthicsApplication)

REC ethics application form (linked 1:1 to a proposal when applicable).

Field	Type	Description
proposalId	ObjectId !' Proposal	Unique sparse index
researcherId	ObjectId !' User	Applicant
status	Enum	draft submitted approved rejected
principal / coResearcher	Object	Person: name, title, faculty, department, phone, email
otherInvestigators	String[]	Up to 6 names
projectTitle, projectLevel	String	Level: undergraduate pgd master
startDate / endDate	Date	Project dates
backgroundLiterature, aimsObjectives, rationale, design	String	Research design
subjectTypes	String[]	human animal records others
inclusionCriteria / exclusionCriteria	String	Participant criteria
risk.level	Enum	no_risk minimal great
instruments	String[]	interview survey observation etc.
consent	Object	hasForm, language (somali english other), items[]
dataSafety / privacy / conflictOfInterest	Object	Data handling & COI
applicantSignature	Object	name, signedAt
approval	Object	decision, certificateId, serialNumber, academicYear, signedByUserId
createdAt / updatedAt	Date	Auto timestamps

References: Proposal, User (researcherId, signedByUserId)

5. projects (Project)

Active research projects created from approved proposals.

Field	Type	Description
proposalId	ObjectId !' Proposal	Source proposal
title	String	Required
researcherId	ObjectId !' User	Principal Investigator (PI)
teamMembers	Array	name, userId, role
milestones	Array	title, dueDate, completed
startDate / endDate	Date	Project timeline
status	Enum	active completed on_hold
progressReports	Array	note, progressPercent, createdBy, createdAt
createdAt / updatedAt	Date	Auto timestamps
References: Proposal, User (researcherId, teamMembers, progressReports.createdBy)		

6. publications (Publication)

Research outputs and faculty publication workflow.

Field	Type	Description
title	String	Required
type	Enum	paper journal_article conference book book_chapter patent thesis review case_study letter_to_editor Bibliographic metadata
year, venue, doi, orcid, url	Mixed	community_research_impact
authors	String[]	Author list
citationCount	Number	Default 0
communityImpact	String	Impact description
status	Enum	draft submitted validated rejected
workflowStage	Enum	submitted in_process pipeline published
researcherId	ObjectId !' User	Owner
validatedBy / validatedAt	ObjectId / Date	Coordinator/director validation
validationComment	String	Review note
createdAt / updatedAt	Date	Auto timestamps
References: User (researcherId, validatedBy)		

7. grants (Grant)

Funding applications and awarded grants.

Field	Type	Description
title, fundingSource, donorRef	String	Grant details
currency	String	Default USD
amountRequested / amountAwarded	Number	Funding amounts
status	Enum	draft submitted approved rejected active closed
complianceNotes	String	Donor compliance notes
researcherId	ObjectId !' User	Applicant
projectId	ObjectId !' Project	Linked project
submittedAt / decidedAt	Date	Workflow dates
createdAt / updatedAt	Date	Auto timestamps
References: User, Project		

8. budgets (Budget)

Budget allocation for grants and projects.

Field	Type	Description
grantId	ObjectId !' Grant	Optional grant link
projectId	ObjectId !' Project	Optional project link
ownerResearcherId	ObjectId !' User	Budget owner
totalAllocated	Number	Total budget cap
currency	String	Default USD
financeNotes	String	Finance officer notes
items[]	Array	type (expense procurement), description, amount, status (pending approved paid rejected)
createdAt / updatedAt	Date	Auto timestamps
References: Grant, Project, User		

9. payments (Payment)

Payment requests against budgets (director !' finance workflow).

Field	Type	Description
category	Enum	research_assistant equipment travel publication_fee other
budgetId	ObjectId !' Budget	Required
payee, purpose	String	Required
amount, currency	Number / String	Payment amount
status	Enum	requested director_approved paid rejected
requestedBy	ObjectId !' User	Requester
directorApprovedBy / directorApprovedAt	ObjectId / Date	Director approval
paidBy / paidAt	ObjectId / Date	Finance disbursement
paymentMethod	Enum	cash bank_transfer mobile_money check other
referenceNumber, notes	String	Payment reference
projectId / grantId	ObjectId	Optional links
createdAt / updatedAt	Date	Auto timestamps
References: Budget, User, Project, Grant		

10. purchaseorders (PurchaseOrder)

Procurement orders against budgets.

Field	Type	Description
poNumber	String	Purchase order number
budgetId	ObjectId !' Budget	Required
vendorName, vendorContact	String	Vendor details
items[]	Array	description, quantity, unitPrice
totalAmount	Number	Auto-calculated from items
status	Enum	requested director_approved paid rejected
projectId / grantId	ObjectId	Optional links
requestedBy	ObjectId !' User	Requester
directorApprovedBy / paidBy	ObjectId !' User	Approvers
paymentMethod	Enum	cash bank_transfer mobile_money check other
createdAt / updatedAt	Date	Auto timestamps

References: Budget, User, Project, Grant

11. researchgroups (ResearchGroup)

Collaboration groups for researchers (join/leave + group chat).

Field	Type	Description
name	String	Required, unique
description	String	Group description
kind	Enum	collaboration thesis
departmentId	ObjectId !' Department	Optional department
createdBy	ObjectId !' User	Creator
members[]	Array	userId, role (lead member), joinedAt
createdAt / updatedAt	Date	Auto timestamps

References: Department, User

12. thesisgroups (ThesisGroup)

Thesis supervision: students, supervisor, meetings.

Field	Type	Description
title	String	Thesis title
students[]	Array	fullName, studentId, email
researchGroupId	ObjectId !'	Linked chat/collaboration group
supervisorId	ResearchGroup ObjectId !' User	Supervising researcher
coordinatorId	ObjectId !' User	Faculty coordinator
department, faculty, facultyResearchArea status	String Enum	Academic context proposed in_progress submitted defended completed
meetingSchedule	String	Recurring schedule text
meetings[]	Array	date, location, agenda, notes, loggedBy
createdBy	ObjectId !' User	Creator
createdAt / updatedAt	Date	Auto timestamps

References: ResearchGroup, User

13. conversations (Conversation)

Direct and group messaging between users.

Field	Type	Description
participants	ObjectId[] !' User	Chat participants
messages[]	Array	senderId, body, at
lastMessageAt	Date	For sorting conversations
groupId	ObjectId !'	Set for group chats
title	ResearchGroup String	Conversation label
createdAt / updatedAt	Date	Auto timestamps

References: User, ResearchGroup

14. notifications (Notification)

Per-user in-app notifications.

Field	Type	Description
userId	ObjectId !' User	Recipient
type	Enum	proposal project grant budget publication repository message system
title, body, link	String	Notification content
readAt	Date	Null = unread
createdAt / updatedAt	Date	Auto timestamps

References: User

15. repositoryitems (RepositoryItem)

Institutional repository file catalog (PDF, CSV, Excel).

Field	Type	Description
type	Enum	dataset publication thesis document
title, description, tags	String	Metadata
filePath, fileSize	String / Number	Stored file
access	Enum	private group institution
groupId	ObjectId !'	Group-scoped access
projectId	ResearchGroup ObjectId !' Project	Project link
uploadedBy	ObjectId !' User	Uploader
createdAt / updatedAt	Date	Auto timestamps
References: ResearchGroup, Project, User		

